Primitive maximal-parabolic reduction for the alternating bound

Human Compression project: L044 working continuation

This section reuses the accepted complete L047 primitive/common theory. Its bounded cases are supplied by the complete required certificate `primitive_certificate.g`, rather than taken from the target source's rank table.

Proposition 1. *Let $G < A_n$ be primitive and have a string C-marking of rank d. For every $n \geq 12$, one has $d \leq \lfloor (n-3)/2 \rfloor$.*

Proof. If the noncommuting diagram is disconnected, the accepted commuting factor lemma identifies $G = T \times T$, with T nonabelian simple and $n = |T| \geq 60$. The intersection property gives $2^d \leq |G| = n^2$, so $d \leq 2\log_2 n < (n-3)/2$. The complete finite argument below settles $12 \leq n \leq 24$; for the rest of this proof assume $n \geq 25$.

For a connected diagram use the exact retained Maróti alternative as in L047, including $A_m^\ell \leq G \leq S_m \wr S_\ell$, $m \geq 5$, and $1 \leq k \leq m/2$ after complementing subsets, where $n = \binom{m}{k}^\ell$. If $\ell = 1$, the case $k = 1$ is excluded by $G < A_n$. Whiston in the faithful m-point action gives $d \leq m-1$. Since $n \geq 12$, the remaining range has $m \geq 6$, and $2(m-1) \leq \binom{m}{2} - 3 \leq n-3$. If $\ell \geq 2$, the same abstract group has the faithful transitive imprimitive action on $m\ell$ letters supplied in L047. Its accepted bound gives $d \leq m\ell/2 + 1 \leq (n-3)/2$, because $m^\ell \geq m\ell + 5$ for $m \geq 5, \ell \geq 2$.

The Mathieu actions all have degree at most24 and belong to the finite boundary. In the remaining alternative put

$$M(n) = n \prod_{j=0}^{\lfloor \log_2 n \rfloor - 1} (n - 2^j).$$

If $d > \lfloor (n-3)/2 \rfloor$, then $2d - 1 \geq n - 3$ and $d \geq 12$ for $n \geq 25$. The retained Conder bound, used only in its valid rank range $d \geq 9$, gives $|G| \geq 2^{2d-1}$. Direct integer multiplication verifies $M(n) < 2^{2(\lfloor (n-3)/2 \rfloor + 1) - 1}$ for $25 \leq n \leq 63$. For $n \geq 64$, write $q = \lfloor \log_2 n \rfloor \geq 6$. Then

$$M(n) < n^{q+1} < 2^{(q+1)^2} \leq 2^{2^q - 3} \leq 2^{n-3},$$

where $(q+1)^2 \leq 2^q - 3$ follows from the case $q = 6$ by induction. This contradicts the lower bound. □

The bounded primitive boundary

For $12 \leq n \leq 24$ the required program runs through every primitive catalogue action contained properly in A_n. The retained completeness of GAP's primitive group library is explicit. Its initial necessary bounds are these:

- an independent d-tuple gives a strict subgroup chain, so $d \leq \Omega(|G|)$, with prime factors counted with multiplicity;
- involutions must generate G;
- an endpoint involution t centralises the other $d-2$ nonneighbouring generators. By IP it is independent of them, so $d \leq 1 + \Omega(|C_G(t)|)$;
- the alternating-index generators form an elementary abelian subgroup of rank $\lceil d/2 \rceil$.

The actual degree12 M_{11} action and degree13 $\mathrm{PSL}_3(3)$ action have elementary abelian two-rank two, verified by all subgroups of a Sylow two-subgroup. This excludes ranks at least five. The actual degree15 action is identified with A_8, so the retained Whiston bound gives rank at most six. For degree24 M_{24}, the involution centraliser orders are exactly $7680, 21504$. At forbidden rank $d \geq 11$ its connected tail of rank $d-2 \geq 9$ satisfies Conder's bound. The endpoint and tail intersect trivially by IP, so its centraliser would have order at least $2 \cdot 2^{2(d-2)-1} = 2^{2d-4} \geq 2^{18}$, a contradiction.

After these necessary tests the exact current GAP catalogue indices are

$$(12,2),$$
$$(16,7),(16,10),(16,11),(16,12),(16,13),(16,15),$$
$$(16,16),(16,17),(16,18),(16,19),(16,20).$$

These are current library indices, not the source's Sims numbering. The program verifies elementary abelian two-rank three for M_{12}, so it need only exclude ranks five and six. Each listed degree16 socle is elementary abelian of order16. Its primitive affine action has irreducible normal module $V = \mathbb{F}_2^4$, and the quotient embeds in $\mathrm{GL}_4(2) \cong A_8$. The accepted L047 normal-module bound and Whiston theorem give $m(G) \leq m(G/V) + 1 \leq 7$. Thus rank seven is the only remaining rank to exclude. No quotient intersection property is assumed.

Lemma 2 (Completeness of the finite prefix search)**.** *The required prefix search excludes every connected string C-marking in the stated rank ranges. It does not use a source-reported maximum rank.*

Proof. List all involutions of the actual permutation group G. Starting with the empty prefix, extend a prefix by every involution which commutes with all earlier entries except its immediate predecessor and does not commute with that predecessor. This is necessary for a connected string diagram. Prefixes are considered up to the normaliser of G in the full symmetric group, then up to the pointwise conjugation stabiliser of the already chosen prefix. These are genuine symmetries of all completion conditions.

Every prefix of a C-marking is independent. It also satisfies intersection of its initial and terminal maximal subgroups equal to the middle subgroup; prefixes failing either test are rejected. These are necessary tests, not an assertion that they alone imply full IP. If a prefix subgroup H has too small an index in G to permit a strict chain of the remaining length, it is rejected. There is one further necessary order test: if the target rank is t and the prefix has length m, the last $t-m-1$ future entries centralise H and generate a group intersecting H trivially by IP. They consequently embed in $C_G(H)/Z(H)$, whose order must be at least 2^{t-m-1}.

A prefix already generating G cannot be extended independently. On complete tuples the program tests all subset intersections against their literal common-index subgroup. Thus every target C-marking is reached, modulo the stated conjugations, unless a necessary condition has already excluded it. The resulting complete lists are empty for all twelve listed actions and rank ranges. A positive control recovers an actual rank-three A_5 marking, so the empty result is not an unconditional output of the engine. □

The complete prefix engine is inherited from the accepted L047 certificate, with only its name and returned report changed. Its new rank ranges and the entire finite primitive screening are part of the present required program. The thirty-nine integer Maróti inequalities for $25 \leq n \leq 63$ are verified there too.

Source scope and retained inputs. This proves the bound asserted in L044 Proposition 2.2 and hence its Corollary 2.3: a primitive maximal parabolic in an A_n C-marking has rank $r-1 \leq (n-3)/2$. Maróti's exact theorem (including its classification dependence), Whiston, Conder in rank at least nine, the accepted L047 common lemmas, and the primitive library and exact GAP group operations remain declared premises. We do not reproduce the source's finer individual maximum-rank claims in Table1 or attribute current GAP catalogue indices to its historical Sims list.